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Research paper

Design and experimental evaluation of a reconfigurable intelligent surface for wireless applications

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ABSTRACT

Reconfigurable Intelligent Surfaces (RIS) have emerged as a transformative technology for enhancing wireless communication beyond 5 G and 6 G networks by dynamically manipulating signal propagation. This paper presents a novel RIS design featuring a circular dumbbell-shaped structure, simulated at a large scale of 32×32 elements, addressing the critical challenge of path loss in mmWave communication. Unlike conventional designs, our approach optimizes phase reconfiguration to enhance beam steering efficiency with a low-cost and efficient design. To validate our design, we conduct an experimental study on a 16×16 phase reconfigurable RIS prototype operating at 32.8 GHz in indoor environments. Each unit cell integrates an AlGaAs PIN diode, enabling 1-bit phase control with a 180° phase shift, ensuring precise wavefront manipulation. Our experimental results confirm the effectiveness of the proposed RIS in beam steering and path loss mitigation, demonstrating its practical feasibility for real-world deployments. This study provides new insights into RIS scalability, design optimization, and high-frequency performance, paving the way for its integration into next-generation wireless communication systems.

1. Introduction

As global 5 G networks expand, the wireless industry is already looking forward to beyond 5 G (B5G) and 6 G technologies, which promise groundbreaking advancements for future applications [1]. Since the commercial rollout of 5 G in 2019, this technology has enabled high-speed connectivity, especially in the Millimeter-Wave (mmWave) bands (24.25 GHz to 52.6 GHz) known as Frequency Range 2 (FR2), which have become essential to the capabilities of IMT-2020 standards [2–4]. The 5 G FR2 has more capabilities to support new mobile application technologies that demand high data rates and aims to increase network capacity by 1000 times with innovative features, such as autonomous systems and remote interaction. The 5 G core has extensive resources for designing technologies like massive Multiple-Input Multiple-Output (m MIMO) [5,6] and mmWave communications [7,8], enabling performance improvements based on the number of antennas that can transmit parallel signals to multiple users using the same frequency-time resource block [9–15]. The primary challenge for high-frequency applications lies in the coverage area of the 5 G FR2 bands, where obstacles in this environment commonly reduce transmitted data due to the sensitivity of high-frequency signals to surfaces, leading to significant path loss. This loss results in diminished received power spatially in dense areas. To maintain the required power for data transmission, more densely placed Base Stations (BS) are frequently necessary to provide adequate signal strength [16,17]. The next-generation 6 G networks, operating in even higher frequency bands such as sub-terahertz to support advanced applications like immersive extended reality and high-quality services, are anticipated to face similar coverage challenges [18,19]. Current 6 G research aims to address these coverage gaps and ensure reliable high-frequency signal transmission.

To address the problem of coverage area, paper [20], introduced one of the most effective techniques. A Reconfigurable Intelligent Surface (RIS) is a new technology that can control how signals travel through the environment. This means that it can create new signal paths when needed, as shown in Fig. 1. Traditionally, wireless networks rely on adjusting the transmission settings of devices to handle changes in the environment, like signal reflections and obstacles [21]. However, these networks can't directly control the environment itself. RIS technology offers a solution by allowing the radio environment to be controlled

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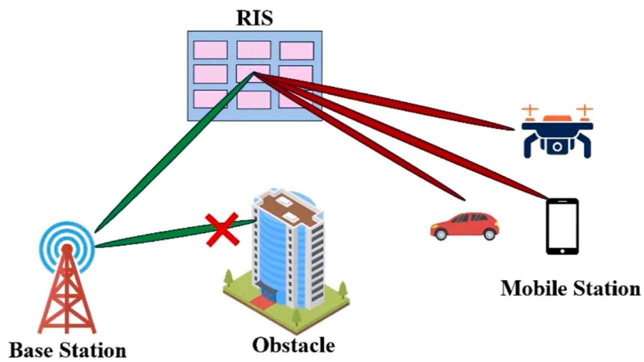


Fig. 1. RIS in the wireless communication system.

through software. When obstacles block the direct path between a BS and a user's device, a RIS can create alternative paths for the signal, reflecting it around the blocked area. RIS is a meta surface made from metal or dielectric materials, composed of many reconfigurable elements called unit cells, arranged in a grid with numerous passive sub-wavelength structures. These structures enable the RIS to reflect signals in specific directions [22,23].

The passive configuration of RIS includes tiny switching devices, such as PIN diodes [24,25], and varactor diodes [26,27] or it can use a tuning material such as graphene [28,29], liquid crystal [30,31], or VO2 [32,33], which allows control over the phase of the reflected signal. Essentially, each unit cell on the RIS can be tuned to modify how it reflects radio signals, adjusting the direction and behavior of the signal as it bounces off the surface. RIS can address the coverage gaps of theoretical analyses in wireless communication [34–39], enhancing outdoor or complex indoor environments [40]. Beyond coverage extension, RIS offers significant benefits in managing interference and reducing signal crosstalk. By dynamically controlling signal reflection paths and adjusting the reflection characteristics (amplitude and phase) directly toward the receiver, it becomes a powerful tool for optimizing network performance as shown in Fig. 1.

RIS has become the essential solution to enhance the 5 G and future 6 G networks, providing a new approach to address the challenges in spatially, mm-Wave applications in indoor environments. These environments are sensitive to physical obstacles, such as smart factories, office buildings, walls, furniture, and even human bodies, which can significantly attenuate the signal, leading to path loss [41]. To ensure conductivity, traditional approaches require denser BS deployments; this solution often achieves a high cost and is logistically challenging. This challenge is extended to 6 G networks operating at higher frequencies to support advanced services. RIS can provide a promising solution to address these challenges. In practical cases, RIS can be mounted on walls or ceilings within indoor settings to control the reflected signal and steer mm-wave signals around obstacles. For example, in a smart building, strategically placed RIS panels can redirect signals to areas that are typically shadowed by physical obstructions, ensuring consistent connectivity for mobile users and critical applications [42]. Similarly, in high-density urban areas, RIS installations on building facades can enhance the coverage of mmWave signals by mitigating the effects of blockage, thereby supporting services such as autonomous vehicle communications and enhanced mobile broadband. The effectiveness of RIS in mitigating high-frequency path loss has been widely analyzed in wireless communication studies [41,42]. RIS has been developed to mitigate interference and improve spatial multiplexing in MIMO systems. However, accurately characterizing RIS-aided communication performance requires a precise mm wave that accounts for real-world conditions [25].

In this work, a novel 1-bit RIS featuring a dumbbell shape has been designed that operates at 32.8 GHz. We validated our approach through extensive simulations on a 32×32 array and experimental tests using a 16×16 prototype, demonstrating its effectiveness in mitigating path loss in indoor environments.

2. Related work

To better represent the mechanics of RIS behavior, some research has sought to enhance these models by incorporating electromagnetic concepts. However, main factors such as (how much each unit cell contributes to the signal and the power received from the RIS) are often missing from current models, which limits their accuracy. Additionally, certain physical effects, such as small phase errors or the way signals reflect directly off surfaces, are sometimes ignored. These factors can lead to inaccurate path-loss predictions, especially considering different distances and angles. To make RIS path-loss models more accurate and useful for real-world applications, it's essential to include these details in the design process [41,42]. In [26], Pei X et al. RIS prototype with 1100 controllable elements operating at 5.8 GHz. The study demonstrated a 26 dB power gain indoors through a 30 cm concrete wall and a 27 dB gain in short-distance outdoor measurements. Long-distance tests successfully transmitted a 32 Mbps data stream over 500 m, enabling smooth 1080p video streaming. With a power consumption of just 1 W, the RIS prototype highlighted its potential for efficient and robust wireless communication systems. Wankai Tang et al. in [42], dissected the role of RISs in enhancing wireless communications. Their work focused on RISs, which are composed of elements that can be adjusted or modified in the unit cells, highlighting their ability to manipulate Electromagnetic Waves (EM) and significantly improve performance and coverage cost-effectively. This study's key contribution was developing free-space path loss models tailored to RIS-assisted wireless communication scenarios. These models were essential for understanding factors such as the distances between the Tx/Rx and the RIS, and the size of the RIS. The experimental results closely aligned with the theoretical predictions, demonstrating the practical enabling of this model in RIS-enabled communication systems. Jungi Jeong et al. proposed an advanced path-loss model for RIS-aided wireless communications [41]. This model incorporates EM considerations, such as the incident and reflected gain patterns of RIS unit cells, the effective received power at the RIS, reflection phase errors, and specular reflection losses. The proposed model is particularly relevant for Beyond 5 G and 6 G networks. A prototype operating at 29 GHz, consisting of 576 phase-reconfigurable unit cells, was developed. Each unit cell utilizes two independently controlled PIN diodes to achieve the desired phase distributions. Experimental results validated the calculated received power, confirming the model's reliability for practical applications. Wankai Tang et al. [43] presented a refined free-space path loss model for RISs, aiming to simplify its application while enhancing its accuracy for RIS-assisted wireless communication systems. The proposed model describes the angle-dependent loss factor, capturing the impact of the directivity of the Tx, Rx, and RIS unit cells on path loss. The study evaluated the properties of individual unit cells, focusing on their scattering performance, power consumption, and physical area. In the context of Radio Frequency Identification (RFID) technology, El-Absi M et al. [44] explored the integration of RISs to enhance real-time remote data capture and connectivity. The authors developed free-space path loss models customed to RIS-assisted RFID communication, incorporating key physical factors such as the Radar Cross-Section (RCS) of tags, the physical properties of the RIS, and the radiative near-field and far-field effects. All these papers propose complex designs that are implemented in the application. The study in [37] developed energy-efficient resource allocation methods and optimized the transmit power and RIS phase shifts while meeting individual link budget requirements. These non-convex optimization problems are addressed using two computationally efficient algorithms: one combining gradient descent for RIS phase design with fractional programming for power allocation, and the other relying on sequential fractional programming. However, the paper does not provide experimental validation of its

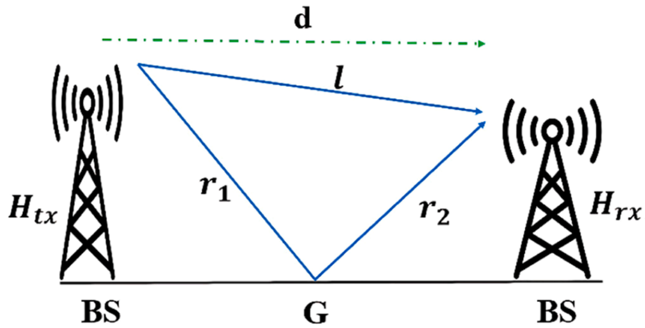


Fig. 2. The Propagation model between two rays showing the direct and reflected paths in a typical urban environment.

findings. In [45], the authors studied resource allocation in wireless networks with distributed RISs. The work focused on maximizing energy efficiency by jointly optimizing transmit beamforming and RIS on-off control under user rate constraints. For the multi-user case, a low-complexity greedy algorithm was proposed. Simulation results demonstrated energy efficiency gains of up to 33 % and 68 % compared to conventional RIS and amplify-and-forward relay schemes, respectively.

This paper provides a comprehensive complete analysis of RIS and addresses path loss challenges in mmWave communication systems. We propose a novel design of dumbbell shape, and a cost-effective RIS unit cell design suitable for mmWave applications. Given the difficulty of controlling phase shifts at high frequencies, we employ spatially oriented PIN diodes capable of operating at mm-wave frequencies. We evaluate the performance of our design under various transmitter angles and demonstrate its effectiveness in improving wireless communication systems by mitigating path loss.

3. Problem formulation

Static meta-surfaces, which have fixed EM functionalities, are widely used in various applications such as radar and satellite systems. However, their limitations in highly dynamic environments become apparent. This path loss severely impacts the received power, making it difficult to maintain reliable communication, especially over long distances or in the presence of obstructions.

The significant path loss that occurs between transmitting and receiving antennas is shown in Fig. 2, the distance between the transmitting and receiving antennas is denoted as l, while the distances between the reflection point G on the ground and the T_x and R_x are represented by r_1 and r_2, respectively. Based on the principles of geometrical optics and Fermat's principle, the G lies along the path that allows the transmitted signal to reach the receiver in the shortest time. This corresponds to the well-known Snell's law of reflection, which states that the angle of incidence (between the incoming ray and the perpendicular to the ground) is equal to the angle of reflection (between the reflected ray and the perpendicular to the ground) [46].

Assuming the transmitted power of a signal x(t) is P_{tx}, the received power P_{rx}, can be expressed as [47]:

$$P_{rx} = P_{tx}\left(\frac{\lambda}{4\pi}\right)\left|\frac{1}{l}+\frac{C\ X\ e^{-j\Delta\Phi}}{r_1+r_2}\right|^{2} \tag{1}$$

Where P_{tx} is power transmitted, $\frac{\lambda}{4\pi}$ represents the free-space path loss factor, accounting for wavelength (λ) and spherical wavefront spreading [47], $\frac{1}{l}$ is the length of the specular reflection path, representing the contribution from the LoS path. $\frac{C\ X\ e^{-j\Delta\Phi}}{r_1+r_2}$ quantifies the contribution of the ground-reflected path, where: C is the complex reflection coefficient, capturing amplitude reduction and phase shift at the ground interface (dependent on material properties and incident angle), $\Delta\Phi = 2\pi\ (r_1 + r_{2-l})/\lambda$ and represents the phase difference between the two paths. The r_1 *and* r_2 are the respective distances between the direct and reflected paths form T_x to reflection point G to R_x. The coherent (phase-sensitive) superposition of both paths means that the signals can interfere constructively or destructively depending on the phase difference. In mmWave applications, the small λ leads to an increase the path loss, which affects the range of communications. To address this challenge, our work introduces a novel solution-based RIS. By integrating the array of the single unit cell, which is designed to steer the signals. Each unit cell adjusts its reflection phase and amplitude, which leads the reflected signals to constructively interfere with the receiver.

In scenarios where the total distance is large enough $d >> H_{tx}$ and H_{rx} (where H_{tx} and H_{rx} is the height of the transmitter and receiver, respectively), the condition simplifies to $l \approx r_1 + r_2$, and the phase difference $\Delta\Phi$ between the two paths becomes negligible. Additionally, for smooth surfaces with minimal scattering, C can be approximated as unity for specular ground reflections ($C \approx 1$). Substituting these approximations into Eq. (1). The equation for P_{rx} then reduces to:

$$P_{rx} = P_{tx}\left(\frac{1}{d^2}\right)^2 \tag{2}$$

which simplifies to a $\frac{1}{d^4}$ power-law dependence. This scaling is characteristic of two-ray ground reflection models, where destructive interference between the direct and reflected paths dominates. This leads to faster signal attenuation compared to free-space propagation.

In free-space environments, in the absence of ground reflections, where the second term in the initial equation is zero, the contribution from the reflected path vanishes. Retaining only the LoS term in Eq. (1) and incorporating antenna gains G_{tx} *and* G_{rx} they received power follows the classical Friis free-space transmission equation:

$$P_{rx} = \ = P_{tx}\ G_{tx}\ G_{rx}\left(\frac{\lambda}{4\pi\ d}\right)^2 \tag{3}$$

Where G_{tx} *and* G_{rx} gain of the transmitting and receiving antennas. In mm-Wave applications, the small λ leads to increased path loss, as evident from the inverse square dependence on d^2. This problem is further exacerbated in NLoS scenarios or when environmental factors like blockages and reflections come into play. Addressing this issue requires innovative solutions to enhance received power and mitigate the adverse effects of path loss.

The concept of programmable metamaterials has emerged, offering dynamic control over the propagation environment by tailoring reflection, refraction, and scattering properties across their surface. These materials can adaptively optimize signal paths, reduce interference, and focus energy on specific directions. This spatial programmability enhances their effectiveness in challenging scenarios like urban environments or NLoS conditions, where high path loss is prevalent. By improving coverage and reliability, programmable materials blur the lines between physical and digital domains, promising to revolutionize wireless network design.

In wireless communication, the received power is a crucial factor influenced by multiple propagation effects, including free-space path loss, specular reflection. When a transmitted signal encounters reflective surfaces such as the ground, the P_{rx} depends on both the direct LoS path and the ground-reflected component. However, these reflections often introduce amplitude attenuation and phase shifts, characterized by the reflection coefficient C and the phase discrepancy $\Delta\Phi$, leading to destructive interference and signal degradation.

To mitigate this issue, we introduce an enhanced received power model based RIS, represents as Z, which accounts for RIS-enabled signal reconfiguration. Unlike the conventional P_{rx} expression, Z is a function of both reflection coefficient and phase shift matrix, where these parameters can be dynamically controlled to maximize constructive interference and improve signal strength. By intelligently adjusting RIS parameters, we can actively shape the wireless channel to counteract

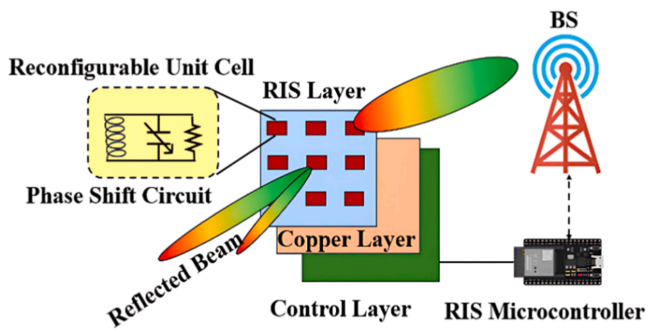


Fig. 3. Schematic diagram for RIS architecture.

path loss, compensate for destructive interference, and enhance overall reception. The following sections provide a mathematical formulation of received power (Z), the role of RIS in controlling reflection coefficient and phase shift variations, and how the introduction of Z leads to improved performance.

4. System model

In this section, we introduce the concept of the received signal in a RIS-aided wireless communication system (Z), which relies on the precise control of amplitude and phase. To establish this foundation, we first present the electric field as a function of amplitude and phase, explaining its role in RIS operation. Additionally, we discuss how phase shifts can be controlled within the RIS structure to optimize signal reflection and improve communication performance.

4.1. Architecture of RIS

In this sub-section, we discuss the basic structure and principles of an RIS. An RIS is generally a flat panel made up of a large number of passive unit cells. These unit cells are designed based on parameters such as arrangement, orientation, shape, and size of the elements, allowing interaction with incident EM waves and reflecting them to the desired user by changing the phase or polarization of the signal. The thickness of an RIS is usually very small, often less than the wavelength of the signal it is intended to control, enabling effective management of signal reflections in a compact form.

In the past, these surfaces were designed to be non-reconfigurable, meaning they were made for a specific use and couldn't be adjusted after fabrication. However, the constantly changing nature of wireless channels, due to user movement and shifting environments, makes a fixed surface difficult to use effectively. Therefore, a reconfigurable surface tuned in real-time is needed to support and enhance wireless communication [48]. As mentioned, electronic components like PIN diodes and varactor diodes are commonly used to adjust the reflection properties of unit cells. These components are popular due to their low cost, fast response times, and minimal signal loss. Additionally, such active elements are widely used in high-frequency applications where path loss or signal strength reduction may occur. These diodes can provide a high phase shift over time, making them suitable for complex environments. In unit cells that use a guided-wave design, an incoming signal from free space is converted into a guided wave within the cell, then reflected and radiated back into the air as a controlled reflected wave.

A general RIS structure is composed of three main layers along with a control unit, as shown in Fig. 3. The top layer is responsible for manipulating incoming EM waves. It comprises reconfigurable elements with a phase shift circuit (resistance, capacitance, and inductance) and conductive patches placed on a dielectric substrate. For practical applications, RF sensors can be added to this layer to detect radio signals and support intelligent RIS configuration, as standard RIS setups cannot directly sense surrounding signals. The middle layer introduces an RF choke to reduce signal leakage and improve isolation between layers. Finally, in the bottom layer, control components are placed to adjust the reflective elements in real time.

These elements are managed by a control system, such as an FPGA, ESP32, or any type of microcontroller, which actively tunes the phase and amplitude of the reflected signals. By precisely configuring these parameters, the control unit enables the RIS to enhance the Z optimizing constructive interference and mitigating destructive effects.

Furthermore, the reflection coefficient and phase shift of the reflected signal can be determined from the electric field $E(\theta,\ \Phi)$ distribution of the RIS, which is influenced by the applied control voltages. By adjusting the local electric field at each RIS element, the system can finely regulate both the reflection amplitude and phase, ensuring optimal wavefront shaping. This amplitude-phase control mechanism plays a critical role in counteracting path loss and maximizing signal strength, as we will describe in detail in the next subsections [49].

4.2. Basic theory of RIS

In a RIS-aided wireless communication system, the received signal can be generally represented as [45]:

$$Z = \left(\sum_{p=1}^{P} \sum_{q=1}^{Q} h_{p,q}^{T_x-RIS-R_x}\ \Gamma_{p,\ q} e^{j\Phi_{pq}} + h^{T_x-R_x} \right) x^{T_x} + v \tag{4}$$

Where, $h_{p,q}^{T_x-RIS-R_x}$, denotes the cascade channel gain from the transmitter to the unit cell (p, q), and the receiver, $h^{T_x-R_x}$ is the direct channel between the transmitter and receiver x^{T_x} is the transmitted signal, and v represents the additive white Gaussian noise.

This equation highlights that Z, the received power, depends on both the amplitude and phase of the reflected signal, which are influenced by the RIS unit cells through the reflection coefficient and phase shift matrix as $|\Gamma_{pq}|e^{i\psi_{pq}}$. To better understand and control these parameters, we derive an equation that explicitly expresses amplitude and phase as functions of the RIS electric field distribution.

In the following, we first introduce the electric field formulation that governs RIS reflections. From this, we establish expressions for phase shifts and amplitude variations, which are essential for optimizing RIS performance and enhancing the

As previously mentioned, when changing the state of each unit cell, the incoming EM waves vary, such as their phase and amplitude. The scattered field from an RIS with $P\ X\ Q$ unit cells for an $x-or\ y-$ polarized incident EM wave can be modeled mathematically as follows [50].

$$E(\theta,\ \Phi) = \sum_{p=0}^{P-1} \sum_{q=0}^{Q-1} B_{pq}\ e^{i\beta_{pq}}.\ |\Gamma_{pq}|e^{i\psi_{pq}}.\ g_{pq}(\theta,\ \Phi)\ \chi\ e^{jk_o\left(pl_x sin\theta cos\Phi + ql_y sin\theta sin\Phi\right)} \tag{5}$$

Where B_{pq} and β_{pq}, denote the relative amplitude and phase concerning each unit cell (p, q), accounting for spatial variations in the incoming EM field, while $|\Gamma_{pq}|$ and ψ_{pq}, are the reflection amplitude and phase of each unit cell. These parameters enable dynamic control over the scattered wave's properties, critical for mitigating path loss Eq. (1)–3. $g_{pq}(\theta,\ \Phi)$, represents the Far-field radiation pattern of the unit cell, defining how the scattered energy is distributed in angular directions θ (elevation) and Φ(azimuth), χ is the Polarization factor, aligning the scattered field with the incident wave's polarization (x or y) polarization. The $e^{jk_o\left(pl_x sin\theta cos\Phi + ql_y sin\theta sin\Phi\right)}$ spatial phase term, capturing the phase progression across the RIS due to the unit cell spacing $l_x,\ l_y$ are the spacings between unit cells along the $x-\ and\ y-$ axes, respectively, and the wavenumber $k_o = \frac{2\pi}{\lambda_o}$. This term ensures coherent beamforming by compensating for path-length differences between unit cells.

By adjusting the reflection amplitude $|\Gamma_{pq}|$, and phase ψ_{pq} from Eq.

(5), We can control each unit cell's scattered EM wave from the RIS, directing it as desired. Assuming the reflection magnitude is consistent across the RIS, we can represent the reflection phase matrix of the RIS as follows [51]:

$$\Theta = \begin{pmatrix} \Phi_{11} & \cdots & \Phi_{1Q} \\ \vdots & \ddots & \vdots \\ \Phi_{P1} & \cdots & \Phi_{PQ} \end{pmatrix} \tag{6}$$

where each element Φ_{pq}, denotes the phase shift applied by the $(p, q)^{th}$ unit cell. Under the assumption of uniform $|\Gamma_{pq}| = constant$, this matrix captures the RIS's ability to dynamically reshape scattered fields by tuning individual phase values Φ_{PQ}.

To achieve optimal phase compensation for each unit cell (p, q), the required phase shift can be calculated as [52]:

$$\Phi_{pq} = K\left(|s - r_{pq}| + |t - r_{pq}| \right) \tag{7}$$

Where, $K = \frac{2\pi}{\lambda}$ is the wavenumber, s is the position of the EM source, t is the target (focusing) point location, and r_{pq} is the position of the unit cell (q, p). The term $|s - r_{pq}|$ represents the distance from the source s to the RIS unit cell r_{pq}, while $|t - r_{pq}|$ is the distance from r_{pq} to the target t. Their sum corresponds to the total propagation path from source to RIS to target and enables adaptive beamforming, where the RIS acts as a programmable lens to concentrate energy at t. This mitigates the $\frac{1}{d^4}$ attenuation form Eq. (2). by compensating for ground-reflection-induced phase discrepancies $\Delta\Phi$ in Eq. (1).

In practical implementations, achieving an ideal continuous phase shift for each unit cell in a RIS is constrained by the discrete nature of phase states available in the hardware [52]. Hence, the phase shifts are often quantized to the closest achievable levels. For instance, in the 1-bit RIS design, the phase control is limited to states, such as 0 or 1. Similarly, in 2-bit, the phase state is limited to four discrete states 0°, 90°, 180°, and 270°, thereby introducing a phase quantization error. This quantization error affects the beamforming accuracy and, subsequently, the performance of the RIS. By increasing the number of quantization bits, the phase approximation can be improved, thereby enhancing the control precision over the reflected wavefront, but at the cost of increased complexity in the unit cell design.

$$\overline{\Phi_{pq}} = \begin{cases} \pi, & if \frac{\pi}{2} \le \Phi_{pq} \le \frac{3\pi}{2} \\ 0, & otherwise \end{cases} \tag{8}$$

Since the reflection response of each RIS unit cell is determined by its phase shift, the quantized phase $\overline{\Phi_{pq}}$ directly affects the way signals are reflected. The total reflection coefficient $\Gamma_{p,\,q}$ is a complex quantity that depends on both the reflection amplitude and the induced phase shift. Given the quantized nature of $\overline{\Phi_{pq}}$, the reflection coefficient can be expressed as:

$$\Gamma_{p,\,q} = |\Gamma_{p,\,q}| e^{j\overline{\Phi_{pq}}} \tag{9}$$

where the magnitude $|\Gamma_{p,\,q}|$ determines the reflection efficiency, and the phase matrix term $e^{j\overline{\Phi_{pq}}}$.

Since the reflection response of each RIS unit cell is determined by its amplitude $|\Gamma_{p,\,q}|$, and phase shift $e^{j\overline{\Phi_{pq}}}$, these factors directly influence the received power. Substituting Eq. (8) and Eq. (9) into the Z in Eq. (4), we observe that RIS elements can be optimized to shape the channel, improving signal strength by controlling both amplitude and phase.

Given the reflection response of each RIS unit cell, we can express the received signal in a more compact form. By representing the RIS-induced reflection coefficients as a vector f and the corresponding channel gains as **h**, the received signal can be rewritten in matrix form as:

$$Z = \left(h^{T} f + h^{T_x - R_x} \right) x + v \tag{10}$$

where h and f, are column vectors of dimension P Q that capture the channel gain vector and reflection coefficient vector, respectively. These vectors are created by vectorizing $h_{p,q}^{T_x-RIS-R_x}$ and $\Gamma_{p,\,q}$. Here, we assume the indices p and q are ordered with p as the major index and q is the minor [53].

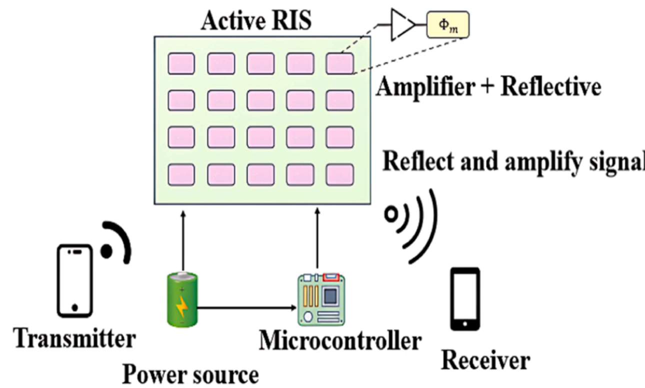


Fig. 4. A scheme of the active RIS.

In scenarios where the direct path between the transmitter and receiver is significantly blocked, the received signal simplifies to:

$$\tilde{z} = h^{T} f + v \tag{11}$$

Where $\tilde{z} = \frac{z}{x}$. In the process of channel estimation, we design R raining patterns with distinct vectors f [54].

By leveraging the structured reflection control of RIS, the received signal can be further generalized across multiple training patterns. Specifically, when designing R distinct training patterns with different reflection coefficient vectors f, we can express the received signals in matrix form as:

$$Z = h^{T} F + v \tag{12}$$

Where Z is a vector of size, F corresponding to each training pattern. This formulation extends the concept of Z from the single-received power expression to a structured set of observations, enabling efficient channel estimation. In high-frequency wireless channels with limited scattering, this problem is often approached as a sparse estimation task, where the objective is to recover the channel gain vector h from multiple measurements of Z.

This final representation reinforces the role of RIS in actively shaping the received power by optimizing both amplitude and phase across unit cells, ultimately enhancing signal reception and mitigating path loss.

4.3. *Active and passive RIS*

The progression from passive to active RIS architecture underscores the diverse trade-offs and applications these technologies enable in modern wireless systems [55].

4.3.1. *Reflection amplifiers RIS*

Active RISs integrate amplification capabilities within the RIS structure as shown in Fig. 4. These architectures enhance the reflection of signal strength, effectively transforming the multiplicative path loss into an additive one. One approach to balance performance and efficiency is to use a single variable gain amplifier with traditional phase-shifting meta-atoms, rather than individual amplifiers for each meta-atom. This design offers a practical solution for active RIS implementation. Unlike passive RISs, active RISs require an external power supply to amplify the signal strength, making it power consumption. Active RIS can adjust their amplification levels and phase shifts to adapt to real-time networks using a control unit that can manage the incidence signal on metamaterial. This approach leads to enhanced SNR. However, implementing them in real-world applications remains complex and

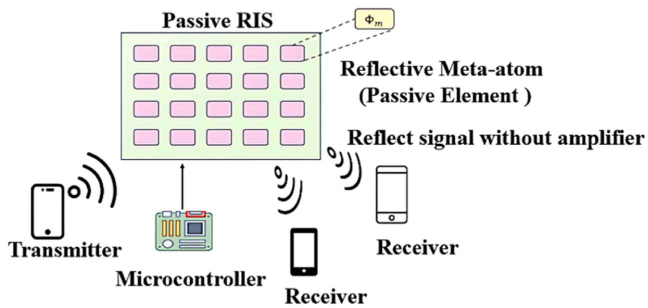


Fig. 5. A scheme of the passive RIS.

challenging [56,57].

Another approach of the active RISs involves amplifying reflect arrays, which reflect amplified scattered fields with a polarization orthogonal to the incident waves. Witch achieved using two-port antennas with orthogonal polarizations, where a standard two-port Power Amplifier (PA) connects the two ports [48,58,59]. Unfortunately, this approach is challenging due to potential polarization limitations that must be addressed to fully unlock their potential.

4.3.2. Reflection RIS

A passive RIS spatial type of intelligent surface, this RIS can manipulate EM waves without needing active power sources or complex signal processing at the surface. It consists of a matrix of sub-wavelength passive reflective elements or "unit cells," which independently modify the phase, amplitude, or polarization of incoming waves. To achieve precise control over the reflecting elements of an RIS, three main approaches have been explored in the literature: (1) Mechanical Actuation, which utilizes mechanisms such as mechanical rotation or translation to physically alter the orientation or position of the elements; (2) Material Selection, which focuses on ensuring that the chosen material's properties are suitable and align with the desired design performance, enabling optimal electromagnetic response; and (3) Tuning Elements, which incorporate active components such as p-i-n diodes or varactor diodes to dynamically control the reflection characteristics of the RIS. Unlike a conventional surface, which reflects incident signals in the specular direction ($\theta_i = \theta_r$) due to a uniform and constant phase delay across the surface, a passive RIS can control phase modulation toward the desired receiver [46].

Fig. 5 illustrates a RIS structure composed of multiple reflective meta-atoms, each capable of dynamically controlling the phase shift Φ_m of its reflective elements. RIS has M reflecting elements to enhance communication between a multi-antenna BS, with N antennas and a single-antenna receiver. Assuming the input signal s(t) is transmitted from the BS and interacts with the RIS. The RIS is represented by the channel matrix $\mathscr{F}^H \in C^{1XN}$, is the channel vector from the RIS to the receiver. The RIS phase shift matrix Φ_m is defined as [60]:

$$\Phi_m = dig(\mathrm{V}) \tag{13}$$

Where $\mathrm{V} = [\, e^{j\theta_1},\ e^{j\theta_2}, \ldots., e^{j\theta_M}]^T$, with $\theta_i \in [0, 2\pi]$ indicating the phase shift applied by the i^{th} RIS element. Each reflecting element contains a PIN diode, which can be switched ON and OFF, allowing a phase difference of π to be introduced [55]. The RIS controller intelligently adjusts the θ_i to align the reflected signals constructively with the LoS component, thereby maximizing the received signal power. The RIS receives the transmitted signal $\mathscr{F}^H$, processes it through the phase shifts, and sends it to the receiver, ensuring an optimal enhancement of the communication performance while maintaining security against potential eavesdroppers [61,62].

In this paper, we focus on passive RIS and provide a comprehensive mathematical analysis of the received power. The theoretical results are simulated and experimentally validated using a spatially optimized RIS design operating in the mmWave frequency range for indoor applications.

5. Received power model based RIS-assisted communication

In high-frequency communication systems, path loss leads to reduce the power of the transmitted signal. RIS has emerged as a promising solution to overcome this challenge. In this section, we develop and analyze the received power model-based RIS-assisted communication. We begin with a model where a single RIS acts as a perfect phase shifter and then extend the analysis to an RIS of multiple reflecting elements.

5.1. Received power model with a single RIS

Consider a similar system model but with an RIS placed on the ground to enhance communication between the Tx and Rx. In this scenario, the RIS acts as a reflective surface that can adjust the direction (reflection angle) and phase of the reflected signal. These adjustments follow the principles of the generalized Snell's law, as outlined in previous studies [63,64]. Specifically, the RIS achieves this by engineering its phase gradient to control the angle and phase of the reflected wave. The reflection coefficient of the RIS depends on various parameters, such as the polarization of the incident EM field, the material properties of the RIS, and the angles of incidence and reflection, as we mentioned. Here, we focus on the RIS's ability to optimize the phase of the reflected signal, assuming no anomalous reflection occurs, and the conventional Snell's law holds.

We assume that the entire ground is covered with an RIS that acts as a perfect phase shifter and the RIS optimally adjusts the phase of the reflected wave to align it coherently with the LoS signal. By aligning the reflected signal's phase with the LoS signal, the total P_r becomes:

$$P_r = P_t \left(\frac{\lambda}{4\pi}\right)^2 \left| \frac{1}{d_1} + \frac{1}{d_2 + d_3} \right|^2 \approx 4P_t \left(\frac{\lambda}{4\pi D}\right)^2 \tag{14}$$

Where d_1 is direct path distance, d_2 and d_3 represent the Tx-RIS and RIS-Rx distance, $D \approx d_1 \approx d_2 + d_3$ is the total effective distance.

This model represents the single configuration of RIS; the reflected signal can combine with the direct signal coherently to enhance the P_r. However, we analyzed the effect of increasing the number of reflecting elements, which leads us to the next mode.

5.2. Received power model with RIS comprising multiple reflecting elements

Now, let us extend the analysis to a more advanced scenario where the ground is not coated with a single RIS but rather an RIS composed of N independently tunable reflecting elements. Each of these elements can control the reflection angle based on Snell's law and adjust the phase of the reflected signal independently of the other elements. For this model, the P_r can be expressed as:

$$P_r = P_t \left(\frac{\lambda}{4\pi}\right)^2 \left| \frac{1}{d_1} + \sum_{K}^{N} \frac{C_k\ e^{-j\theta_k}}{d_{1,k} + d_{2,k}} \right|^2 \tag{15}$$

Where k is k^{th} reflecting element. To maximize the P_r, we assume that each C_k is optimized such that the reflected signals from all N reflecting elements align coherently with the LoS signal. Mathematically, this requires $C_i = e^{j\theta_k}$ and $d_{1,k} + d_{2,k} \approx d_1$ for all k. Under this condition, the P_r simplifies to:

$$P_r = (N+1)^2\ P_t \left(\frac{\lambda}{4\pi D}\right)^2 \tag{16}$$

Eq. (16) shows that the P_r scales quadratically with the number of reflecting elements, increasing N, which improves the P_r strength.

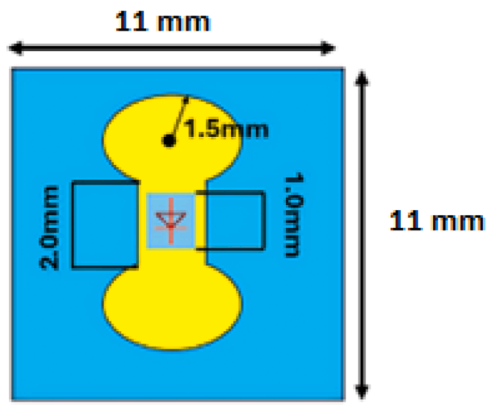


Fig. 6. A single-unit cell structure of the proposed RIS.

Table 1
The proposed RIS unit cell specification.

Specification	Value
Operating frequency (GHz)	32.8
Polarization	Linear
Layers	One Layer
Application	mm-wave
Bandwidth	30–33 GHz
Dimension(mmˆ3)	11×11×0.2
Substrate	RO4003C
Pin diode	MADP 000,907–14020P-AlGaAs

5.3. *Received power model with RIS comprising multiple reflecting elements*

From Eq. (16), it can understand the required size of an RIS can be understood as a function of the number of reflecting meta-surfaces N. As a reference, we consider meta-surface examples available in [65,66]. For instance, based on [65], a single meta-surface capable of controlling the angle and phase of a reflected signal typically has a size of approximately 10 cm x 10cm. If we assume an RIS is composed of $N = 100$ independently controllable reflecting meta-surfaces, the total size of the RIS would be approximately 100 cm x 100cm. At an operating frequency of around 30 GHz and λ around 1cm. According to Eq. (16), the P_r scales proportionally to N^2, resulting in a gain of 100^2 or 10^4 which corresponds to an increase of approximately 40 dB compared to conventional reflection setups.

The model presented above analyzes the basis for understanding how RIS can mitigate path loss. In the next section, we will explore the design of our design, which can be implemented in high-frequency communication systems, to validate the theoretical discussion.

6. Proposed design of RIS

In this section, we present our proposed design of the RIS. We discussed the unit cell design, including specifications for both amplitude and phase reflection at both states of the PIN diode.

6.1. *Unit cell design*

Fig. 6 shows the simulated design of a single-unit cell in the proposed RIS. The unit cell consists of a single layer, effectively reducing signal loss. At high frequencies, signals traveling through multiple layers can experience increased losses due to additional material and dielectric interfaces. A single-layer design minimizes transitions and potential signal attenuation, which is essential for maintaining performance at mmWave frequencies as we discussed in detail in [67].

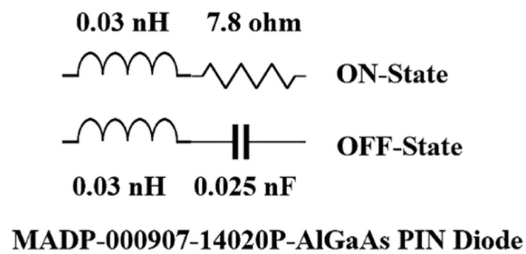
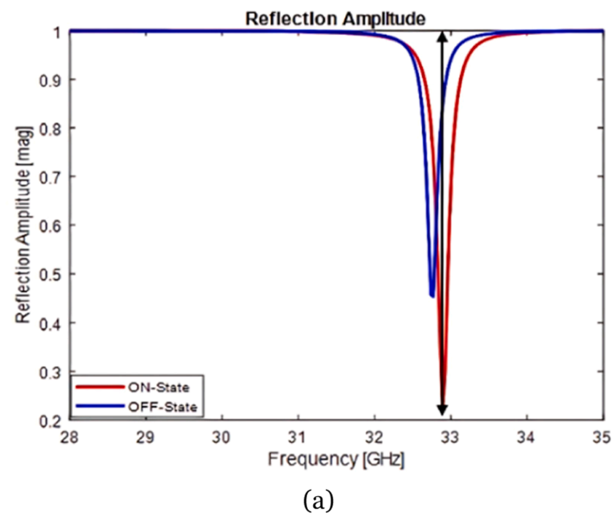


Fig. 7. Equivalent RLC model for AlGaAs PIN diode for both ON/OFF states.



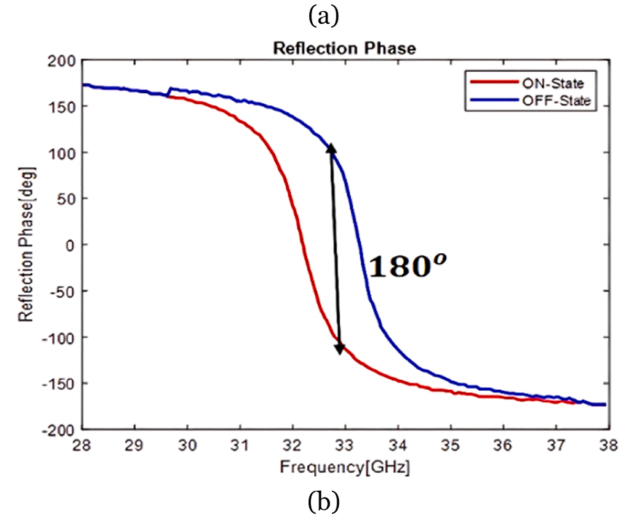


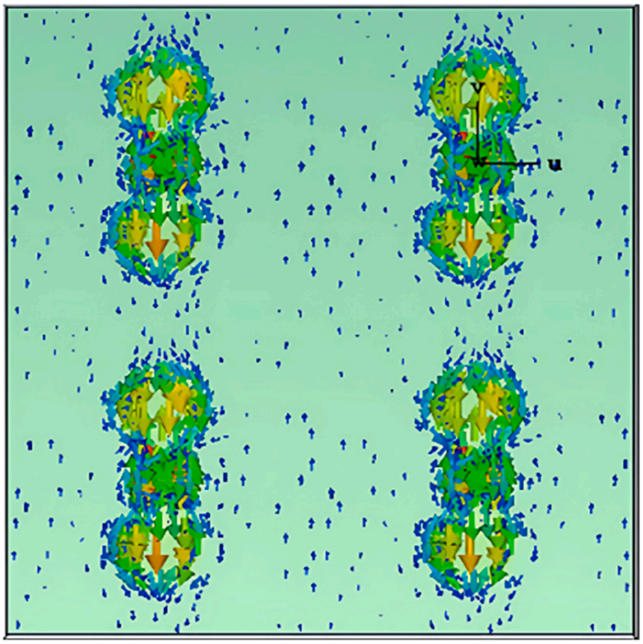
Fig. 8. Reflection characteristics of the proposed single unit cell (a) amplitude, (b) phase.

6.1.1. *Design Specifications*

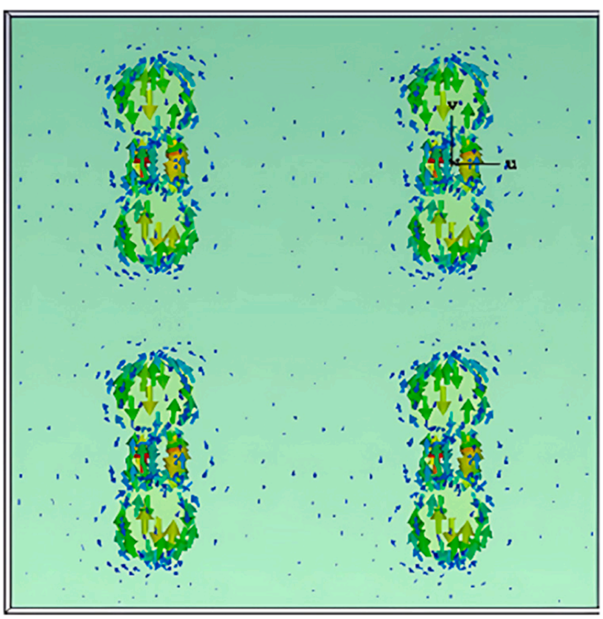
Table 1 presents the specifications of the proposed design of the RIS. The Rogers RO4003C substrate was used, with a relative permittivity of 3.38. The unit cell dimensions are 11×11 mm². Periodic boundary conditions were applied to simulate an infinite array configuration, and 'Floquet' ports were used to analyze the structure's reflection and transmission characteristics accurately. This setup ensures that the simulated results are representative of the RIS array performance in real-world applications.

6.1.2. *Tunning element*

A PIN diode MADP 000,907–14020P-AlGaAs is integrated into each element of the unit cell to control the phase of the incident wave. This



(a)



(b)

Fig. 9. Surface current distribution of the proposed unit cell (a) ON, (b) OFF states.

type of diode is suitable for mm-wave applications due to its minimal parasitic capacitance and inductance, which ensure fast switching between states and enable control of the reflection phase. It minimizes signal degradation and supports high-frequency operation with low insertion loss.

As shown in Fig. 7. When the PIN diode is in the ON state, it behaves as a series combination of resistance and inductance. Conversely, in the OFF state, it behaves as a series combination of capacitance and inductance. Thus, under EM wave radiation, the impedance of the PIN diode can be expressed as (Z_L):

$$\begin{cases} Z_{ON} = R + jwL \text{ ON state} \\ Z_{OFF} = \dfrac{1}{jwC} + jwL \text{ OFF state} \end{cases} \tag{17}$$

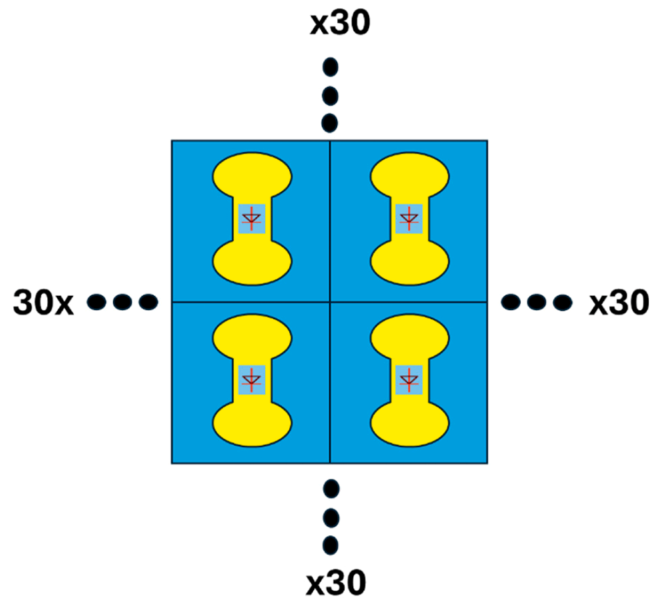


Fig. 10. Proposed design of 32 RIS arrays.

The reflection coefficient is then determined by:

$$\Gamma = \frac{Z_L - Z_R}{Z_L + Z_R} = |\Gamma| e^{j\theta} \tag{18}$$

Where Z_R represents the radiation impedance of the unit cell. The unit cell structure is specifically designed to achieve the desired value of Z_R. Enabling 180° phase shift between the ON and OFF states at the operating frequency. As a design configuration is at 32.8 GHz, the design achieves a 180° phase shift at this frequency. Fig. 8, illustrates the reflection characteristics of the single unit cell in the on and off states at the center frequency of 32.8 GHz, (a) represents the amplitude for and (b) the refection phase at both states.

6.1.3. *Reflection characteristics*

In this subsection, we analyze the reflection characteristics of the designed unit cell, specifically focusing on the amplitude and phase responses at the center frequency of 32.8 GHz. The unit cell's reflection properties are evaluated in both ON-state and OFF-state conditions.

6.1.4. *Surface current distribution*

The surface current density of the proposed design at both resonant frequencies is illustrated in Fig. 9. In (a) the ON-state, the current surface exhibits a maximum distribution concentrated near the AlGaAs region of the single-layered structure. In contrast, the surface current (b) shows a lower magnitude in the OFF state, indicating reduced activity. The resonance behavior is further validated through the surface current distribution along the structure.

6.2. *An 1-bit RIS design for large-area wavefront control*

In this subsection, we introduce the scaling of the RIS to a 32×32 array that introduces advancements in manipulating EM waves over larger areas, enabling precise control of the reflected amplitude and phase.

The simulation was extended to a 32 × 32 array configuration, resulting in a total surface size of 563 × 563 mm^2. The unit cell spacing exceeds the typical half-wavelength criterion (0.5λ) for preventing grating lobes. Therefore, the spacing of 1.2λ may introduce additional lobes in the far-field pattern depending on the beamforming configuration. The objective was to analyze the behavior of the EM wave when incident on the expanded RIS, particularly under normal incidence

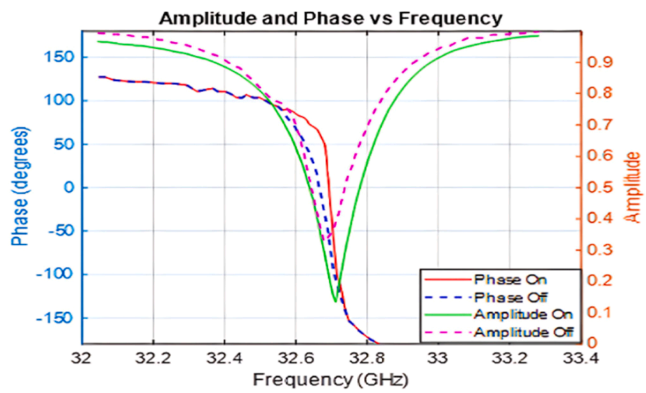


Fig. 11. Reflection characteristics (amplitude and phase) V.S. frequency of the proposed 32 RIS.

Table 2
RIS array specification.

Specification	Value
Array size	563×563 mnˆ2
Wavelength λ	9.15 mm
Array size in λ	61.56 λ x 61.56 λ
Spacing element	1.2 λ
No. of array	32×32
No. of diodes	1024 pin °1 bit
Bits	**180** ° **(0,** 180°)
Phase difference	

conditions (where the $\theta_i = \varphi_i = 0$). The simulation focused on evaluating the phase and amplitude control across all unit cells. A total of 1024 PIN diodes were incorporated into the design to enhance both the resolution and control granularity of the RIS, enabling precise manipulation of the reflected wavefront as shown in Fig. 10.

Fig. 11 represents the reflection characteristics of the proposed 32 RIS. The reflection amplitude for the ON state exhibits a sharp and narrow dip at the resonance frequency of approximately 32.6 GHz, where the amplitude decreases to below 0.2. This indicates strong resonance behavior, demonstrating the RIS's ability to effectively attenuate or amplify specific frequency components. In contrast, the OFF state shows a broader and less pronounced dip, with higher amplitude values across the frequency range. The increased number of unit cells in the 32×32 array amplifies the collective response of the surface, creating a more defined and sharper resonance for the ON state.

The reflection phase curves reveal a distinct difference between the ON and OFF states. In the ON state, the phase transitions smoothly and sharply around the resonance frequency, achieving a phase shift of approximately 180° across the range. This steep phase transition indicates precise control over the phase modulation. The OFF state, on the other hand, exhibits a more gradual and less pronounced phase variation. The larger array size enhances the phase resolution due to the increased number of elements contributing to the reflected wavefront, enabling finer control over the phase distribution and thus more effective wavefront shaping. Furthermore, our design spacing of the unit cell is set at 1.2λ. To increase this spacing up with this spacing could lead to other issues, such as the production of some additional grating lobes and increased mutual coupling effects in the far field.

This scaling of our proposed 32×32 RIS design increases the area of coverage, which aims to reduce the path loss effect by refining the phase control even when the receiver is moving. Our design demonstrates a highly efficient control of the reflecting signal, with a resonance peak around 32.8 GHz and effective amplitude and phase modulation. However, expanding this design to larger arrays, such as 64×64 or 128×128 arrays, introduces additional challenges. For illustration, while we increase the number of unit cells to improve the beamforming,

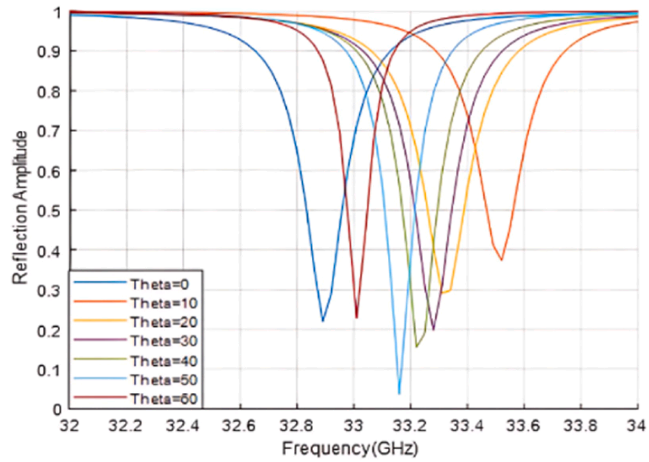


Fig. 12. Reflection amplitude under deferent θ_i from $0^o to\ 60^o$ and $\varphi_i = 0$.

Table 3
List the real-world constraints.

Constraint	Specific Challenges	Potential Solutions/Mitigations
Manufacturing Costs	High raw material and labor expenses	- Optimize design and manufacturing processes. - Use cost-effective materials
Environmental Factors	Exposure to harsh weather, temperature extremes, and contaminants	- Employ robust materials and protective coatings - Conduct environmental testing
Complexity	Compatibility issues	- Develop modular and flexible designs - Implement phased testing and integration

that leads to an increase in the complexity of the control circuit and produces signal leakage due to the need to manage a higher density of PIN diodes. Additionally, a large array requires more power distribution and improved thermal management to support the higher power consumption.

Table 2 provides detailed specifications of the design parameters. To validate the reflection characteristics of the expanded array, simulations were conducted at the center frequency of 32.8 GHz under two conditions: when all elements are ON and when all elements are OFF.

Fig. 12 represents the reflection amplitude at different θ_i from $0^o to\ 60^o$and $\varphi_i = 0$, demonstrating the angular behavior of the design. However, due to limitations in prototype development, we discuss and analyze a scaled-down 16 × 16 array in the following section.

6.3. Addressing real-world constraints

Implementing solutions in settings requires consideration of various real-world constraints that can affect feasibility. Table 3 summarizes the main constraints and their challenges, and we suggest possible strategies and solutions for mitigation during our study. Manufacturing costs often include expenses for materials and equipment. To address this, our design contains Rogers material, which is suited for attenuating signals in high-frequency applications. Environmental factors, such as temperature fluctuations and humidity are significant challenges in real-world applications. Finally, complexity is a critical factor we consider in our design. To ensure simplicity, we implemented a single-layer RIS structure with one PIN diode per unit cell. The diode was chosen for its ability to maintain precise tuning control within the desired frequency range. Additionally, the circular dumbbell-shaped resonator design enables 1-bit phase control in this range.

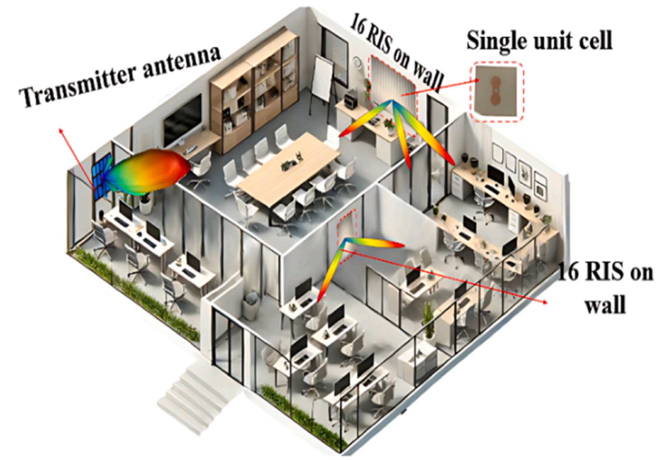


Fig. 13. Schem diagram of the proposed RIS design in indoor applications.

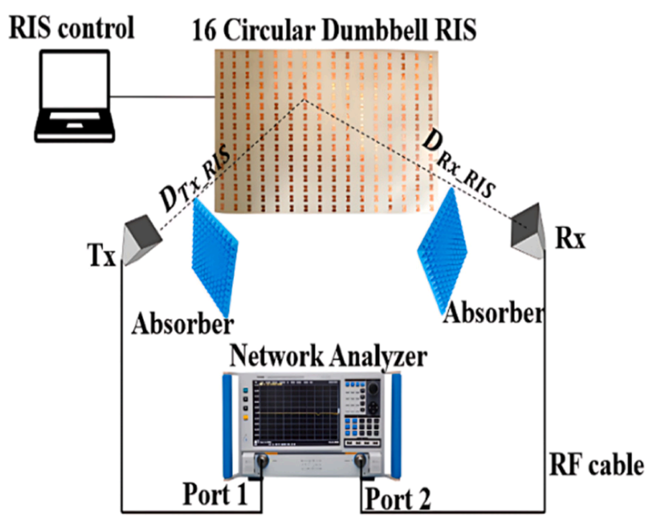


Fig. 14. Measurement setup for proposed 16 RIS.

7. Fabrication and measurement

In this section, we present the measurement characteristics of the fabricated our RIS to validate the proposed path loss model in indoor communication environments. The objective is to demonstrate how our design enhances wireless 5 G communication systems by improving the power received in practical scenarios, such as conference room setup. Fig. 13 illustrates the deployment of the RIS in an indoor environment, where the RIS panels are strategically placed on the walls to optimize signal reflections. The transmitter is positioned at one end of the room, and the signal is reflected by the RIS to ensure uniform coverage and improved signal strength across the space.

7.1. Measurement setup

The measurement setup for evaluating the P_r between the Tx and Rx is illustrated in Fig. 14. A 16×16 RIS is placed between the Tx and Rx to reflect the transmitted signal. The Tx transmits a Continuous Wave (CW) signal toward the RIS, and the reflected signal is received at the Rx. A Vector Network Analyzer (VNA) is employed to measure the S-parameters, particularly S_{21}, which corresponds to the transmitted power from Port 1 (Tx) to Port 2 (Rx), enabling the calculation of the received power. RF absorbers are strategically placed around the setup to minimize unwanted reflections and multipath propagation, ensuring that the measured signal predominantly corresponds to the RIS-reflected path.

Table 4
Measurement specification.

Setup	Specification
Tx an°rn antennas	LB–28–10-A
Tx an°in	10 °to 40 GHz
Tx an°° range	38 cm
Distance from Tx to RIS	35 cm
Distance from RIS to Rx	16×16
RIS array	256
No. of unit cell	176 mm x176 mm
Demotion of RIS	From 10 to 60 step 10
Rx scan angles	From 30 to 60 °
Tx scan angles	

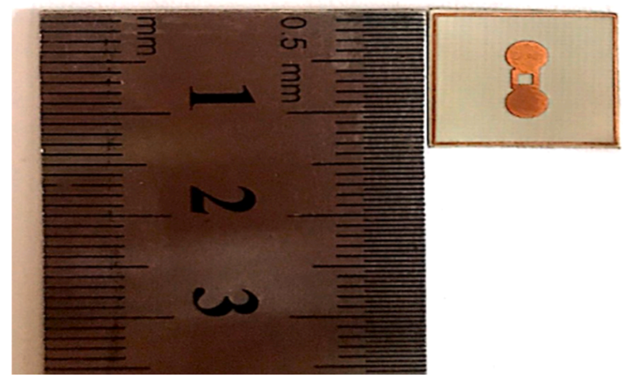


Fig. 15. Fabricated design of the proposed unit cell of the circular dumbbell shape.

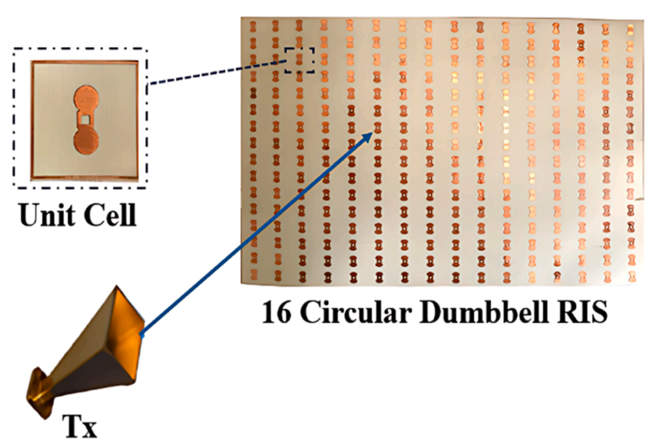


Fig. 16. A proposed 16×16 Circular Dumbbell RIS.

Table 4 describes details the specification of the measurement setup. And Fig. 15 and Fig. 16, shows the prototype single RIS and 16×16 array, receptively, under proposed setup. To measure the reflection beam, the Rx horn antenna is rotated from 10° to 60° while maintaining a fixed distance of 35 cm from the RIS.

7.2. Beam scanning methode

To measure the received power for different angles of reflection, a beam scanning method is employed, as illustrated in Fig. 17. In this method, the Tx horn antenna is fixed at a specific incident angle with respect to the RIS, while the Rx horn antenna is systematically repositioned to capture the reflected beam at multiple angles. The measurement begins by fixing the Tx horn antenna at an incident angle, typically starting at 30° and increasing incrementally to 60°

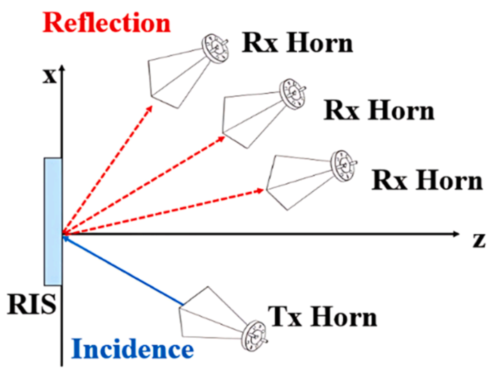


Fig. 17. Schematic representation of RIS-assisted wave reflection, where an incident signal from the Tx is redirected towards multiple Rx.

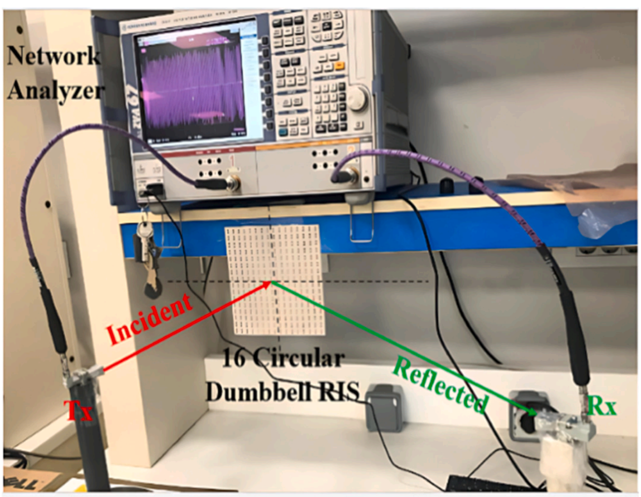


Fig. 18. Measurement setup of proposed 16 Circular Dumbbell RIS in a real indoor application environment.

At each fixed Tx angle, the Rx horn antenna is scanned across various reflection angles. This scanning is performed by physically moving the Rx horn along a predefined trajectory, allowing it to capture the power of the reflected signal from the RIS at each position. The reflected beam is observed and measured at discrete angular steps, and the corresponding received power is recorded using a network analyzer.

When the RIS is in the off state, it behaves like a Perfect Electric Conductor (PEC), providing a standard reflection phase. Therefore, we first measure the characteristics of the proposed RIS in its off state to evaluate its baseline reflection characteristics, which serve as a reference for comparison with its active state. This approach ensures that the beam is reflected without any active modulation, enabling us to study the RIS's behavior in terms of controlling the beam direction.

7.3. Measurement result

Fig. 18 represents the measurements setup of the proposed 16 RIS. This setup demonstrates the characterization of a RIS to measure the P_r under controlled conditions. The RIS prototype, placed between the Tx and Rx horn antennas, facilitates beam reflection to redirect the signal path. A VNA is used to measure the power at the receiver by analyzing the S-parameters, with the incident and reflected signals clearly marked.

The measurement setup was conducted outside the chamber to evaluate the behavior of the RIS in a real indoor application environment. The initial measurement setup was configured with the incidence angles $\theta_i = \varphi_i = 0$ at the center of the RIS. The Tx horn antenna was then systematically repositioned clockwise from 30° to 60° with a discrete step size of 10. For each Tx position, the reflected power was captured by repositioning the Rx horn antenna at corresponding angular steps 10 from 30° to 60° This process was performed to validate the beam scanning approach and assess the performance of the RIS in steering the reflected beam.

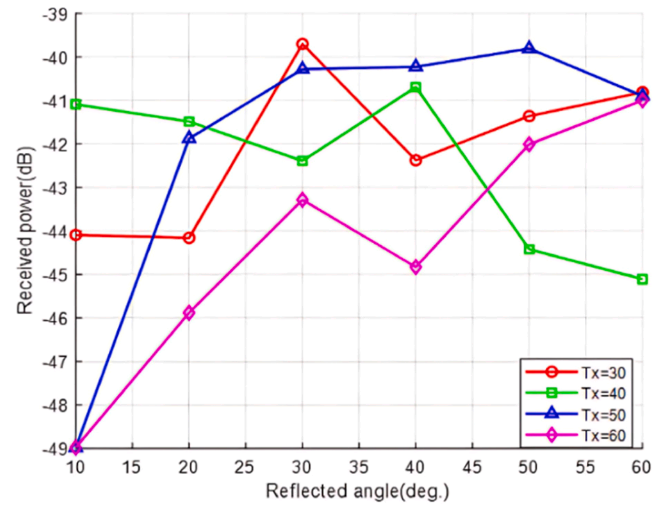


Fig. 19. Received power under the different incident angles and moving receiver.

Several challenges could have produced limitations in our experimental validation. First, fabrication tolerances led to the introduction of minor variations in the phase response and reflection coefficients of the single element, such as variation during the fabrication of the unit cell. Measurement setup is a critical factor in ensuring accurate results, as it relates to minimizing noise or interference when evaluating the performance of the proposed RIS under varying incident angles. For indoor applications, our measurements are conducted in an anechoic chamber to isolate the RIS's true performance from external disturbances. However, as we mentioned, the environment's factors such as temperature and humidity variations may effect on our setup, potentially influencing measurement accuracy.

7.4. Received power measured

Fig. 19 presents the P_r as a function of the reflection angle for

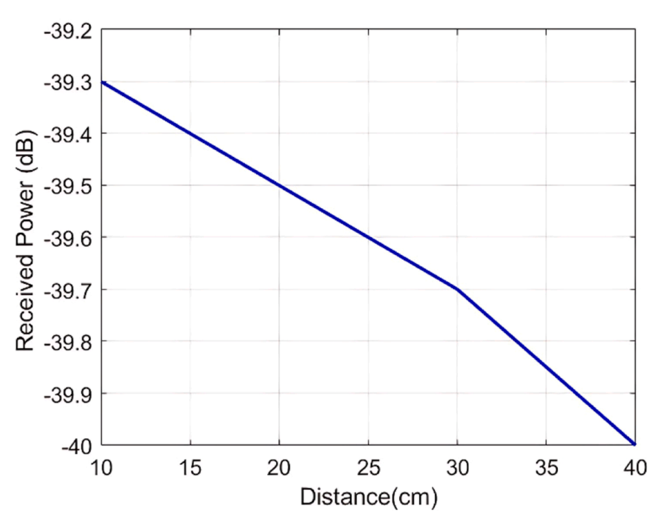


Fig. 20. Measured received power as a function of distance, demonstrating path loss behavior and validating RIS performance.

Table 5
It Compares the implemented RIS with a different study.

Ref.	Freq.	Unit Cell Shape	phase	Layer /Control	NO. of diode in a unit cell	Simulation/ Experiment	Key Contributions
[26]	5.8 GHz	Uniform Square RIS	1-bit	3 layers/ Varactor diode	2 Diodes	Fabricate 55×20 RIS	Development and experimental validation of an 1100-element RIS prototype
	29GH	Metasurface-based RIS	2-bit	6 layers/ PIN diode	2 Diodes	Fabricate 24×24 RIS	Improved phase loss
[41]	10.5 GHz	Metasurface-based RIS	-	1 layer	-	Fabricate 100×102 RIS 50×34 RIS 8 × 32 RIS	Improved phase loss
[62]	32.5GHz 33.6GHz	RIS with four symmetric circle sector	1-bit	1 layer/ PIN Diode (AlGaAs)	1 Diode	Simulate 3 × 3RIS	Novel design for mmWave
[25]	28.5GHz	Rectangular waveguide (WR-34)	1-bit	1 layer/ PIN Diode	1 Diode	Fabricate 20×20	Novel design for mmWave
This work	32.8GHz	Dumbbell-shaped Unit Cells	1-bit	1 layer/ PIN Diode (AlGaAs)	1 Diode	Simulation (32×32) & Experiment (16×16)	Large-scale design, validated for real-world path loss mitigation for indoor application

different fixed $\theta_i = 30^o, 40^o, 50^o$ *and* 60^o. This choice was made to streamline the process of beam steering and ensure a systematic alignment of the reflected wavefront. By arranging the elements in this manner, we simplify the control logic required for phase adjustments. For Tx = 30°, the P_r reaches its peak around a reflected angle of 30°, demonstrating effective beam alignment. Similarly, for Tx = 40° and 50°, there are distinct peaks at different reflected angles, indicating optimized reflection for these configurations. However, for Tx = 60°, the received power is consistently lower across all reflected angles, suggesting reduced alignment efficiency. These results highlight the RIS's capability to manipulate the reflected beam direction to maximize received power and its sensitivity to transmitter positioning. The observed P_r values confirm that the RIS can achieve significant reflection gains, validating the proposed design.

The P_r measurements obtained from the experimental setup validate the performance of the RIS, also confirm the correctness and applicability of the theoretically received power model. The measured and theoretical results demonstrate the effectiveness of the RIS in achieving significant power gains through beam steering. Furthermore, any discrepancies observed between the theoretical predictions and the measured values can be analyzed in detail to refine the model or account for practical factors such as hardware imperfections, alignment errors, or signal losses.

7.5. Received power variation with fixed RIS-to-Moving receiver distance

To further evaluate the proposed RIS measurements, we measured P_r at varying distances from the fixed RIS to the moving receiver. Fig. 20 illustrates the P_r as a function of distance, which shows power attenuation as we increase the separation distance. The results indicate that the decrease in P_r, aligns with the expected path loss model. These measurements validate RIS's ability to maintain signal strength over extended distances. Additionally, the consistency between the measured and theoretical values further supports the efficiency of our RIS, making it suitable for mm-wave applications. This validation highlights the feasibility of the proposed design for real-world applications, particularly in indoor communication environments. Additionally, this design will be even more impactful in the next development phase, where the RIS can be made intelligent. By integrating real-time control mechanisms and advanced algorithms, the RIS will be able to dynamically interact with EM waves, eliminating the need to measure and adjust all elements manually. In comparison to existing RIS designs, this work presents a novel approach that balances cost-effectiveness and high efficiency. Unlike previous studies that employed complex multi-layer structures or a higher number of diodes per unit cell, our design utilizes a single-layer configuration with AlGaAs PIN diodes in a dumbbell-shaped unit cell at 32.8 GHz. This minimizes fabrication complexity and significantly reduces costs while maintaining efficient phase control, as represented in Table 5. Additionally, unlike prior works that focused solely on simulations or smaller-scale experiments, this study advances the field by implementing a large-scale RIS design, validating both simulation results 32×32 and experimental measurements 16×16 in the indoor environment.

The high efficiency observed in real-world measurements demonstrates the potential of this RIS for practical indoor applications, effectively mitigating path loss while maintaining a low-cost structure compared to other high-frequency RIS implementations.

8. Future work

In future work, we plan to extend our circular dumbbell RIS optimization. Especially focusing on the optimization of RIS phase configurations and transmit beamforming. This optimization mainly incorporates machine-learning techniques such as reinforcement learning to enable dynamic RIS in real-world environments. Additionally, this optimization technique will address uncertainties and challenges while ensuring scalability for large-scale applications. Ultimately, our goal is to connect the gap between theoretical and practical deployments, which enhance the coverage and energy efficiency of the 6G.

Conclusion

This paper investigated the design, simulation, and experimental validation of a novel Reconfigurable Intelligent surface (RIS) for mmWave communication systems, addressing the critical challenge of path loss in indoor environments. A 32×32 dumbbell-shaped RIS was simulated using a CST simulator with periodic boundary conditions at 32.8 GHz, showcasing a unique large-scale unit cell design optimized for high-frequency operation. To validate the practical feasibility of our approach, a 16×16 phase reconfigurable RIS prototype has been fabricated and tested. The experimental results confirm that the 1-bit phase control, enabled by AlGaAs PIN diodes, achieves a precise 180° phase shift, enhancing beam steering, improving received power, and effectively mitigating path loss in indoor environments.

CRediT authorship contribution statement

Haider TH.Salim ALRikabi: Writing – original draft, Software, Resources, Methodology. **Adheed H. Sallomi:** Supervision. **Hasan F. KHazaal:** Supervision, Writing – review & editing. **Ahmed Magdy:** Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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